MITHILESH A

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LinkedIn | GitHub | StackExchange

PROFESSIONAL SUMMARY

High-achieving AI & Data Science student at SASTRA University specializing in Reinforcement Learning, Deep Learning, and Quantitative Finance. Proven track record of engineering profitable algorithmic trading models (52%+ net return) and optimizing heavy machine learning models for constrained edge hardware. Adept at bridging advanced mathematical research with scalable, real-world software architecture.

EDUCATION

SASTRA Deemed University — Thanjavur, TN
B.Tech in Computer Science (AI & Data Science) — *Aug. 2024 – June 2028*

- **Core Concepts:** Data Structures & Algorithms (DSA), OOP, DBMS, Multi-Agent Systems, High-Performance Computing.

Harvard University (CS50) — Remote
Web Programming with Python and JavaScript (CS50W) — *In Progress*

PROFESSIONAL & RESEARCH EXPERIENCE

WorldQuant — Remote
BRAIN Research Consultant (Prev. Conditional Consultant) — *Apr. 2026 – Present*

- Engineered logic-based quantitative alphas and probability-driven financial signal strategies using Custom Expression Languages.
- Applied statistical modeling to construct, evaluate, and perfectly backtest alpha ideas across global equities.

XYlofy Ai — Bengaluru, India (Remote)
AI & Data Science Intern — *Jun. 2026 – Jul. 2026*

- Developed and deployed scalable machine learning architectures for real-world automation projects, collaborating with industry mentors to optimize data pipelines.

SASTRA Deemed University — Thanjavur, TN
Undergraduate Researcher — *Feb. 2026 – Present*

- **Quantitative Finance (IEEE Paper Submitted):** Authored a paper on predicting USD/INR exchange rates using a novel Spatial-Temporal Proximal Policy Architecture (ST-PPA).Achieved **95%+ predictive accuracy** perfectly backtested; generated a **52.94% net return** (Sharpe Ratio 0.560).
- **Cybersecurity (APT Detection):** Investigated threat detection methodologies by leveraging Deep Learning models to analyze network traffic and dynamically identify cyber vulnerabilities.

TECHNICAL PROJECTS

RL-Based Traffic Management (SIH 2025) — Sep. 2025 – Dec. 2025

- **Tech Stack:** DQN, MARL, OpenCV, C++, Raspberry Pi, Python. Architected a Multi-Agent Reinforcement Learning (MARL) control system to minimize intersection clearance times. Optimized computationally heavy OpenCV vehicle classification algorithms for low-latency execution on constrained edge hardware.

Web-Based AI Applications — Jan. 2026 – Present

- **Tech Stack:** Python, Django, React, MongoDB, Machine Learning.Engineered full-stack web platforms integrating custom ML backends and REST APIs for data visualization.

* For more projects visit github profile and check out the portfolio website.

TECHNICAL SKILLS

- **AI/ML & Deep Learning:** Reinforcement Learning (PPO, DQN, MARL), Neural Networks, Policy Gradient, TensorFlow, Backpropagation, Transfer Learning, Model Optimization, Time Series.
- **NLP & Computer Vision:** LLMs (Transformers, BERT, GPT, RAG), Prompt Engineering, Text Classification, Embeddings, OpenCV, Object Detection, Image Segmentation, CNNs, YOLO.
- **Programming:** Python (Advanced), C++, Java, C, SQL. **Web Stack (Learning):** JavaScript, Django, React, HTML/CSS, MongoDB, REST APIs, Flask.
- **Data, Quant & Systems:** NumPy, Pandas, Matplotlib, Feature Engineering, EDA, Algorithmic Trading, Exchange Rate Modeling, Risk Analysis, HPC, Git, Linux, Raspberry Pi, VS Code.

CERTIFICATIONS & ACHIEVEMENTS

- **Certifications:** Yale (*Financial Markets*), NPTEL (*Deep Learning*), Cisco & Kaggle (*Data Science & ML*).
- **CMI STEMS & Math Olympiad:** Selected in the **Top 30 in India** for research aptitude; Merit certificate holder (IOQM). Achieved Gold Genius Level - WQB for consistent and profitable quantitative alpha generation.